



HAM



VS

SPAM



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Brief Marketing

Les utilisateurs d'AT&T sont confrontés à l'afflux constant de spams via SMS. Le signalement manuel est lent, inefficace à grande échelle et incapable de réagir en temps réel.

Afin d'améliorer l'expérience utilisateur et de renforcer la sécurité, AT&T souhaite **automatiser la détection des spams** à partir du contenu des messages texte.

Les livrables



Un modèle simple CNN / LSTM ou GRU



Un modèle de Transfert Learning type BERT



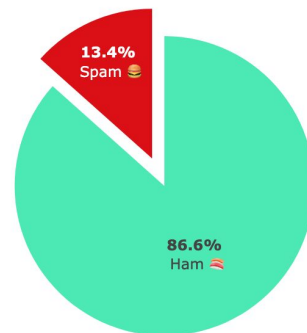
EDA

NA 99,1%, 99,78%, 99,89

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN

	v1	count	Target
0	ham	4825	Ham 🍖
1	spam	747	Spam 🍔

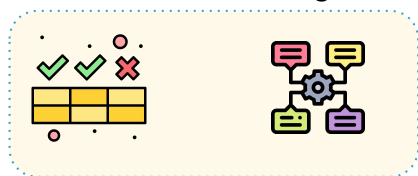
Répartition entre **Ham** 🍖 et **Spam** 🍔





Modèle CNN

Data Cleansing



Token Speciaux
(features
engineering) :

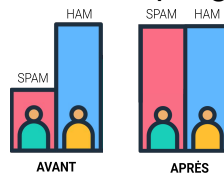
<URL>
<MONEY>
<CLIC>

etc.

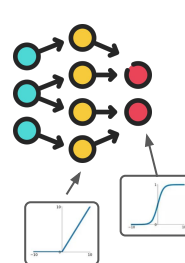
Tokenizing



OverSampling



CNN



Output

85%

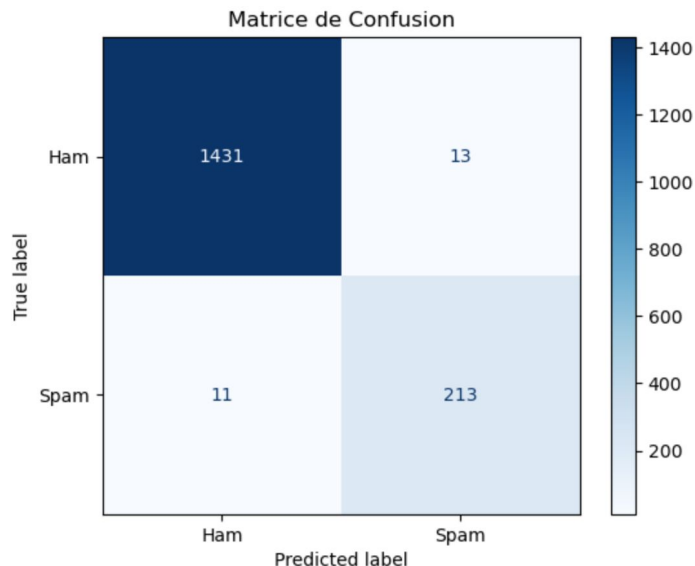
15%



Résultats CNN

Vrais positifs

Faux positifs



Faux Négatifs

Vrais Négatifs

Classification Report:				
	precision	recall	f1-score	support
Ham	0.99	0.99	0.99	1444
Spam	0.94	0.95	0.95	224
accuracy			0.99	1668
macro avg	0.97	0.97	0.97	1668
weighted avg	0.99	0.99	0.99	1668

94% des prédictions
sont correctes

95% correctement
identifiés

Pour un modèle créé de toute pièce, il se débrouille plutôt bien.



Tests

SMS	Prédiction	Score de confiance	Réel
"Hey Marty, we have a meeting tomorrow, don't forget it",	 HAM	99,74%	 HAM
You have won a \$1000 Walmart gift card! Click here to claim now!",	 SPAM	99,54%	 SPAM
"Reminder: Your appointment with Dr. Smith is at 3 PM tomorrow.",	 HAM	56,17%	 HAM
"URGENT: Your bank account has been compromised. Verify your details immediately.",	 SPAM	97,15%	 SPAM
"Urgent: your banck account is compromised. Click here: http://bit.ly/xxx ",	 SPAM	96,87%	 SPAM
"You have 1 unread message. Reply to unlock: bit.ly/msg123 ",	 SPAM	95,60%	 SPAM
"Security alert: suspicious login from NY. Confirm now: verify-link.com ",	 HAM	63,58%	 SPAM
"FREE MONEY! CLAIM YOUR PRIZE NOW! NO LIMIT!",	 SPAM	94,40%	 SPAM
"Hello, can we meet tomorrow?",	 HAM	99,78%	 HAM
"Congratulation, you've won an iphone, clic here now to grab it!",	 SPAM	79,56%	 SPAM
"Hello, do we have a meeting today ? Could you call me?",	 HAM	98,35%	 HAM
"Your just won 500 buck, claim it now, clic here !",	 SPAM	97,20%	 SPAM
"Charles, let's work together on this projet.",	 HAM	99,67%	 HAM
"Hello Aline, why don't you answer to my call ?",	 HAM	95,99%	 HAM
"Eric, could you schedule a meeting for next week.",	 HAM	86,25%	 HAM
"Maybe the false negatif is to high. We should improve it.",	 HAM	99,53%	 HAM
"We are please to announce that you have won an iphone 16, click here : http://shor.lu/122 "	 SPAM	98,24%	 SPAM
"We offer you 10 grands for you effort. It's a not a big deal for us. Thank you",	 HAM	60,05%	 SPAM
"Your verification code is 123456. Do not share it with anyone.",	 SPAM	72,87%	 HAM
"Hello Eric, i have your answer, call me asap."	 HAM	93,63%	 HAM

17/20, le modèle tient le choc sur des SMS fabriqués pour l'occasion

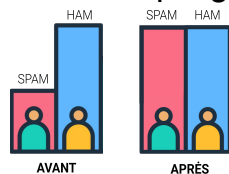


BERT (distilbert base uncased)

Data Cleansing



OverSampling



AutoTokenizing
(distilbert-base-uncased)



DistilBERT



SoftMax

Output

85%

15%

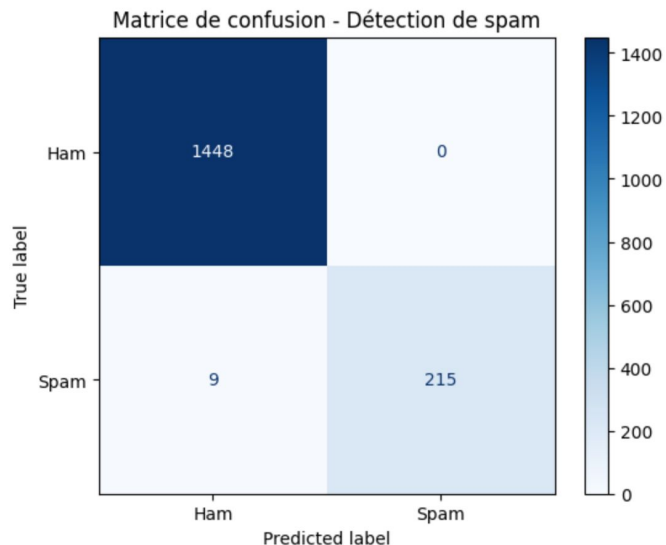
Bert : Bidirectional encoder representations from transformers
distilbert-base-uncased : modèle allégé (66 millions de paramètres versus 110 millions) avec la base de Bert et ne tenant pas compte de la casse.



Résultats BERT

Vrais positifs

Faux positifs



Faux Négatifs

Vrais Négatifs

Rapport de classification :

	precision	recall	f1-score	support
Ham	0.99	1.00	1.00	1448
Spam	1.00	0.96	0.98	224
accuracy			0.99	1672
macro avg	1.00	0.98	0.99	1672
weighted avg	0.99	0.99	0.99	1672

100% des prédictions
sont correctes

96% correctement
identifiés

Bert remporte le match !



Tests

SMS	Prédiction	Score de confiance	Réal
"Hey Marty, we have a meeting tomorrow, don't forget it",	 HAM	100,00%	 HAM
You have won a \$1000 Walmart gift card! Click here to claim now!",	 SPAM	100,00%	 SPAM
"Reminder: Your appointment with Dr. Smith is at 3 PM tomorrow.",	 HAM	100,00%	 HAM
"URGENT: Your bank account has been compromised. Verify your details immediately.",	 HAM	98,74%	 SPAM
"Urgent: your banck account is compromised. Click here: http://bit.ly/xxx ",	 SPAM	99,99%	 SPAM
"You have 1 unread message. Reply to unlock: bit.ly/msg123 ",	 SPAM	100,00%	 SPAM
"Security alert: suspicious login from NY. Confirm now: verify-link.com ",	 SPAM	98,27%	 SPAM
"FREE MONEY! CLAIM YOUR PRIZE NOW! NO LIMIT!",	 SPAM	99,99%	 SPAM
"Hello, can we meet tomorrow?",	 HAM	100,00%	 HAM
"Congratulation, you've won an iphone, clic here now to grab it!",	 SPAM	99,99%	 SPAM
"Hello, do we have a meeting today ? Could you call me?",	 HAM	99,99%	 HAM
"Your just won 500 buck, claim it now, clic here !",	 HAM	93,46%	 SPAM
"Charles, let's work together on this projet.",	 HAM	100,00%	 HAM
"Hello Aline, why don't you answer to my call ?",	 HAM	99,99%	 HAM
"Eric, could you schedule a meeting for next week.",	 HAM	100,00%	 HAM
"Maybe the false negatif is to high. We should improve it.",	 HAM	100,00%	 HAM
"We are please to announce that you have won an iphone 16, click here : http://shor.lu/122 "	 SPAM	100,00%	 SPAM
"We offer you 10 grands for you effort. It's a not a big deal for us. Thank you",	 HAM	100,00%	 SPAM
"Your verification code is 123456. Do not share it with anyone.",	 SPAM	97,67%	 HAM
"Hello Eric, i have your answer, call me asap."	 HAM	100,00%	 HAM

16/20, grosse surprise, avec des métriques supérieures, Bert fait un peu moins bien. Il est probable qu'il s'est calé sur le vocabulaire d'avant.

Des prédictions qui sont en "overconfident".



Conclusion

La puissance de BERT est impressionnante mais :

- Problème : ressource computationnelle ainsi que d'inférence.
- Les petits modèles plus "légers" peuvent challenger ce type de classification.

Warning : Notons que le DataSet date déjà. Le vocabulaire, les techniques de Spam ont changé et surtout les SMS de Spam sont corrélés à l'actualité (exemple en France : impôts, vacances, énergies etc.).

Next Step :

- Enrichir le DataSet des SMS d'aujourd'hui. Créer plus de [Token Speciaux](#).
- Créer une interface pour que les clients puissent contribuer à l'enrichissement du DataSet
- Comme AT&T fait déjà un classement manuel, prévoir que ce classement viennent enrichir aussi le dataset.
- Faire une veille systématique afin d'anticiper les Spam d'opportunité.



**Merci de
votre attention**